Project Planning & Management Document

1. Project Overview

- *Elevator Project!*
- *Samruddhi Mullerpatan & Kavya Karkhanis*

2. Team Roles & Responsibilities

Member Name	Role	Responsibilities
Samruddhi	Software, documentation, testing and planning, mechanical design	Thonny and MIT app inventor codes iterations, mechanical design of pulley and structure, testing planning of system, documentation of process and making of video and ppt
Kavya	Hardware and circuitry, mechanical design, software testing	Circuit design and planning, mechanical design of servo doors and sensors, Thonny and MIT app inventor code testing, documentation of process.

3. Project Timeline & Milestones

Week	Goals	Deliverables	Status
Week 0	Initial prototyping, figuring out door mechanism + structure	Sketches of structure, initial making of sliding doors.	Partially completed
Week 1	Converting rotatory movement to linear for servo motors, making pulley system, making	3d printing of rack and pinion, pulley wheels, and sourcing of pulley cable, build building structure, servo code	Completed

	of building structure, writing of code for servos		
Week 2	Finalizing pulley system mechanism, IR sensor placement and working, DC motor code and IR code	Make pulley structure with dc motor, stick IR sensors to floors of building, DC motor code and IR code	Partially completed
Week 3	Work pulley with elevator car and counterweight, finalize IR working, assembly of all components together, work on MIT app inventor and Bluetooth code, franken code with all components	Assembly together with wiring iterations, breadboard placement iterations, MIT app inventor blocks and Bluetooth code, Franken code	Completed

Break project into weekly achievable goals

4. Weekly Progress Log

Week 1

- Planned Tasks: Initial prototyping, figuring out door mechanism + structure
- Completed Tasks: Sketches of structure- a few completed, initial making of working sliding doors, servo code
- Challenges Faced: Issues with dimensions of sliding doors for smooth movement, converting servo's rotatory movement to linear for said doors, making of pulley wheel models on Fusion360
- Solutions: Research and advice from Nayan on servo movement mechanism- rack and pinion, troubleshooting for 3d printing model making from peers
- Next Steps: Ensuring that servos work together smoothly, doors open properly, wiring is concealed in all applicable places, making of the reference building to figure out dc motor code.

Week 2

- Planned Tasks: Converting rotatory movement to linear for servo motors, making pulley system, making of building structure, writing of separate codes for dc motor movement
- Completed Tasks: 3d printing of rack and pinion, pulley wheels, and sourcing of pulley cable, building structure, dc code
- Challenges Faced: cable of pulley was slack, counterweight of incorrect weight and at inappropriate height
- Solutions: Secondary research and advice on fixing cable material and tightening it, ensuring the placement and weight of the counterweight is correct
- Next Steps: Ensuring that dc motor pulley works well in terms of going up and down movements, working of all components together in sync, working of IR sensors with proper timing

Week 3

- Planned Tasks: Work pulley with elevator car and counterweight, finalize IR working, assembly of all components together, work on MIT app inventor and Bluetooth code
- Completed Tasks: Assembly together with wiring iterations, breadboard placement iterations, MIT app inventor blocks and Bluetooth code, IR code, complete franken code
- Challenges Faced: cable groove width of pulley too narrow thus cable kept getting stuck and falling off during elevator car movement, counterweight kept hitting car during movement, resetting of motor movement in code
- Solutions: Used our backup pulley wheel for a wider groove, added a diverter pulley in the shaft to solve counterweight obstruction issue, use of interrupts in code for motor movement across floors
- Next Steps: Working on visuals, concealing wiring, making the project aesthetically appealing and clean in terms of finishing.

👉 Encourage honesty here—failures are important learning